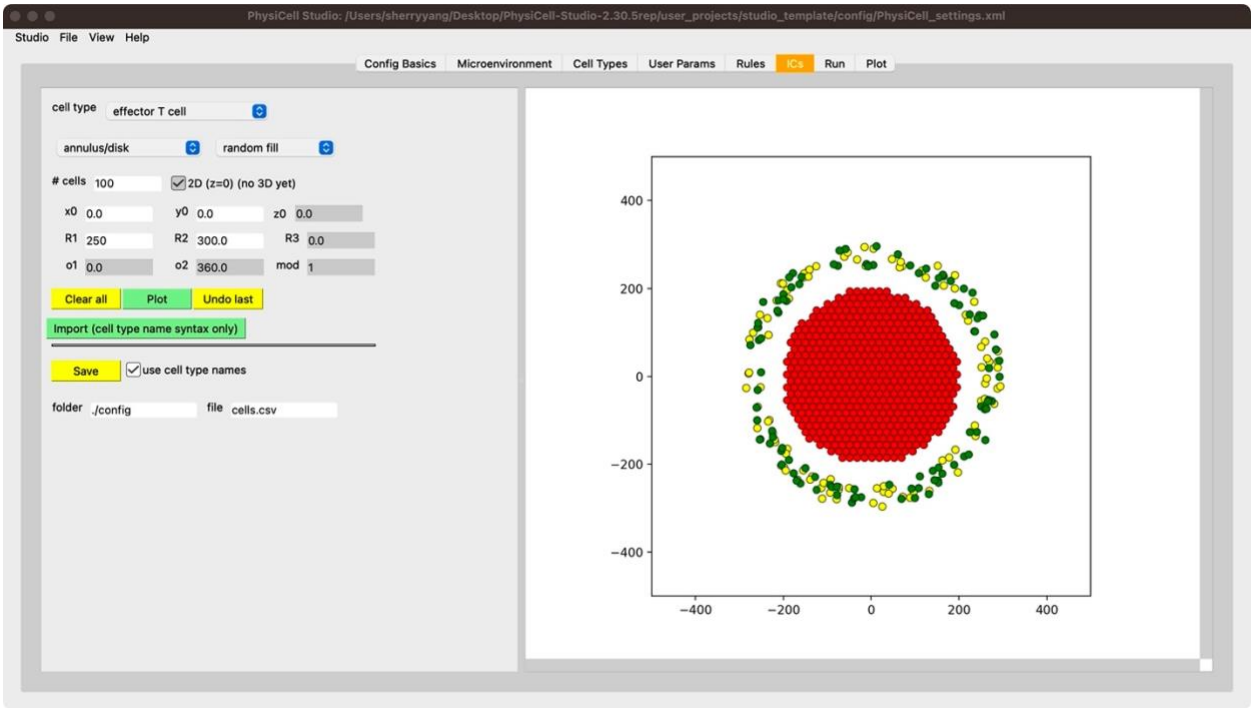
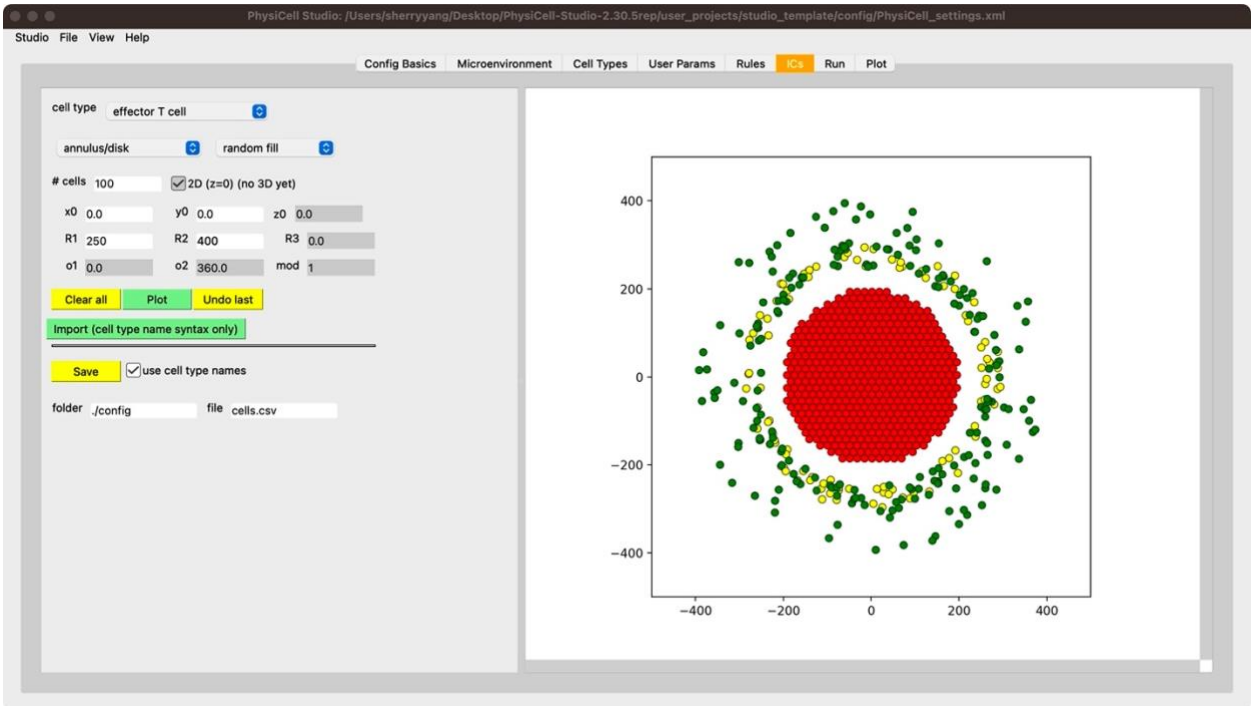
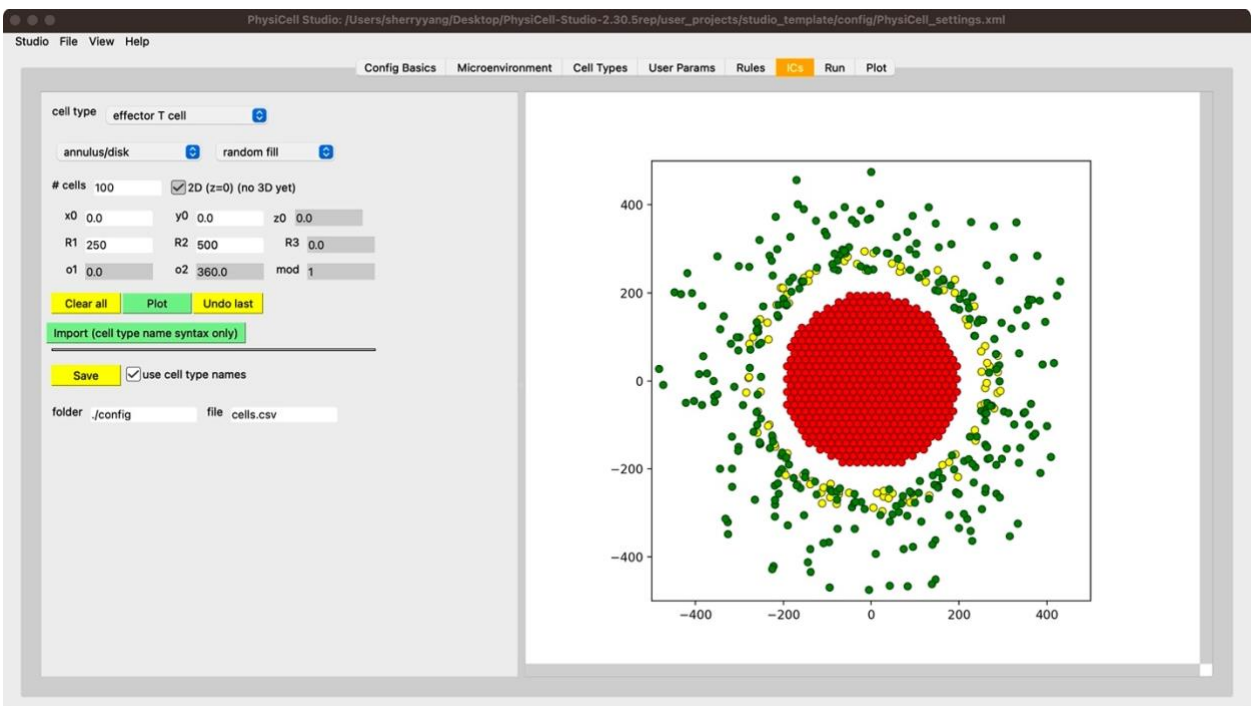
Control Case effector T cell

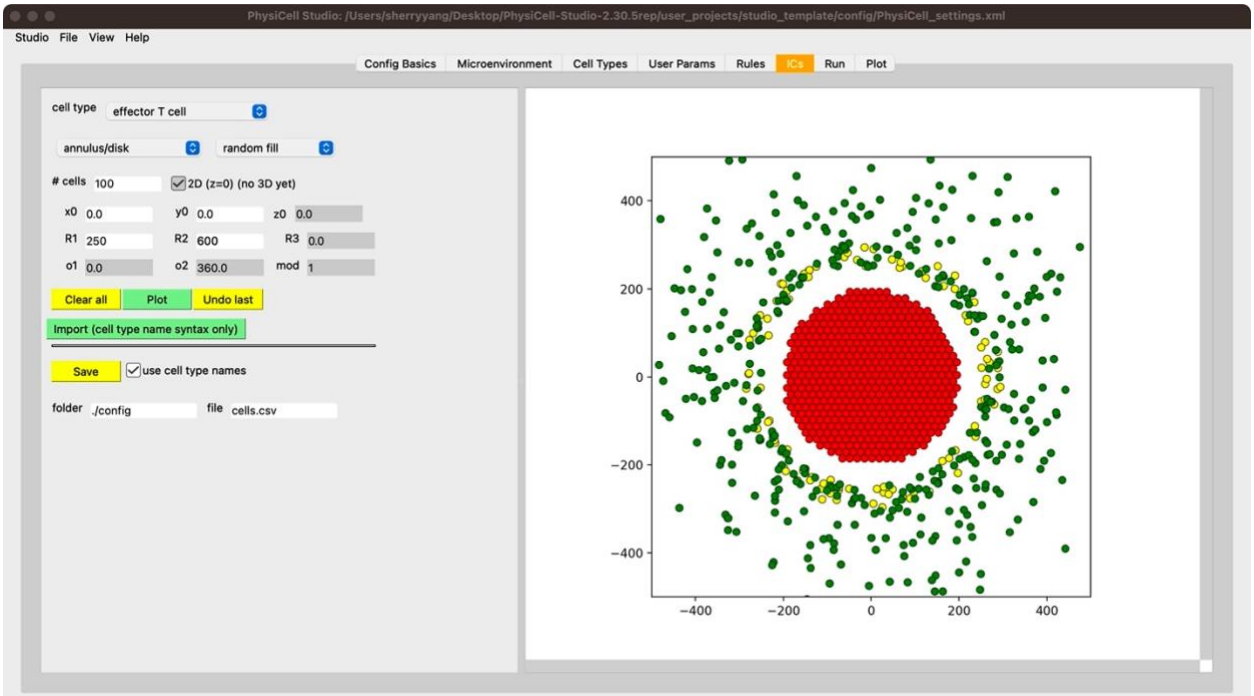


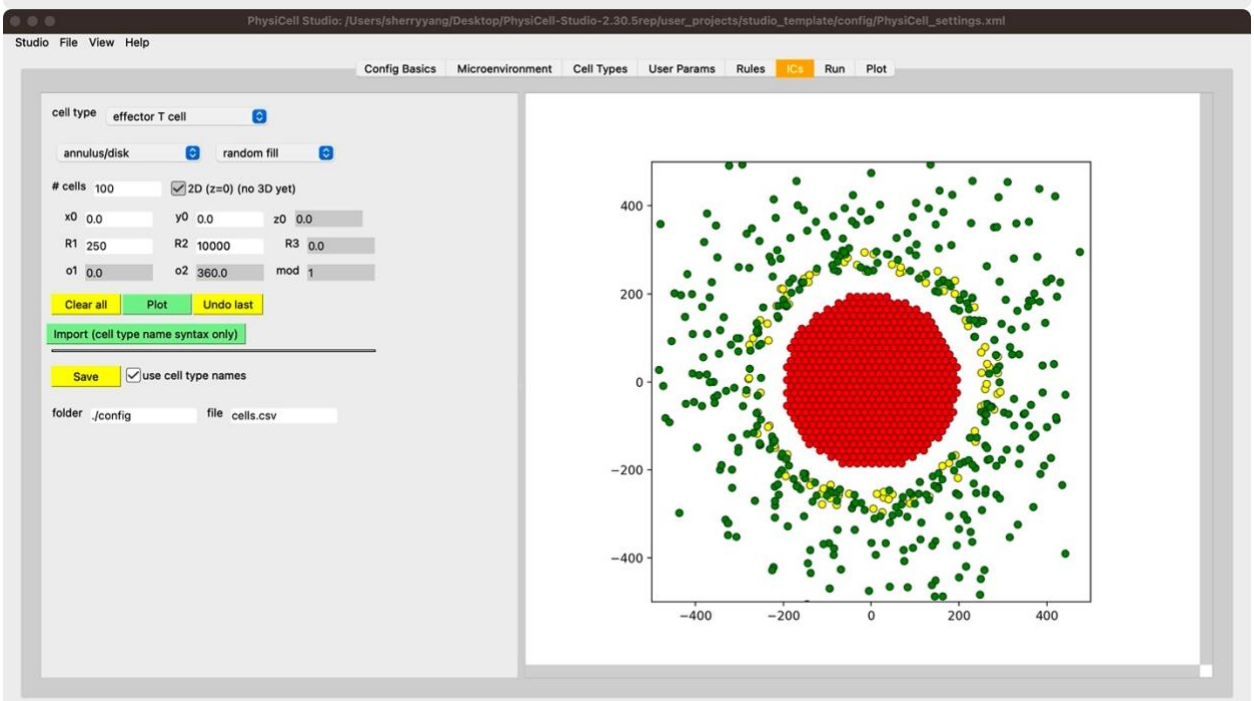
Cancer cell in red surrounded by macrophages and effector T cell on the outer circle

Now change R2 parameter, which is the outer circle radius, by setting values to 400, 500, 600, 10000


PhysiCell Studio: /Users/sherryyang/Desktop/PhysiCell-Studio-2.30.5rep/user_projects/studio_template/config/PhysiCell_settings.xml
Studio File View Help
Config Basics Microenvironment Cell Types User Params Rules ICs Run Plot
cell type effector T cell
annulus/disk random fill
cells 100 2D (z=0) (no 3D yet)
x0 0.0 y0 0.0 z0 0.0
R1 250 R2 400 R3 0.0
o1 0.0 o2 360.0 mod 1
Clear all Plot Undo last
Import (cell type name syntax only)
Save use cell type names
folder ./config file cells.csv
400
200
0
−200
−400
−400 −200 0 200 400

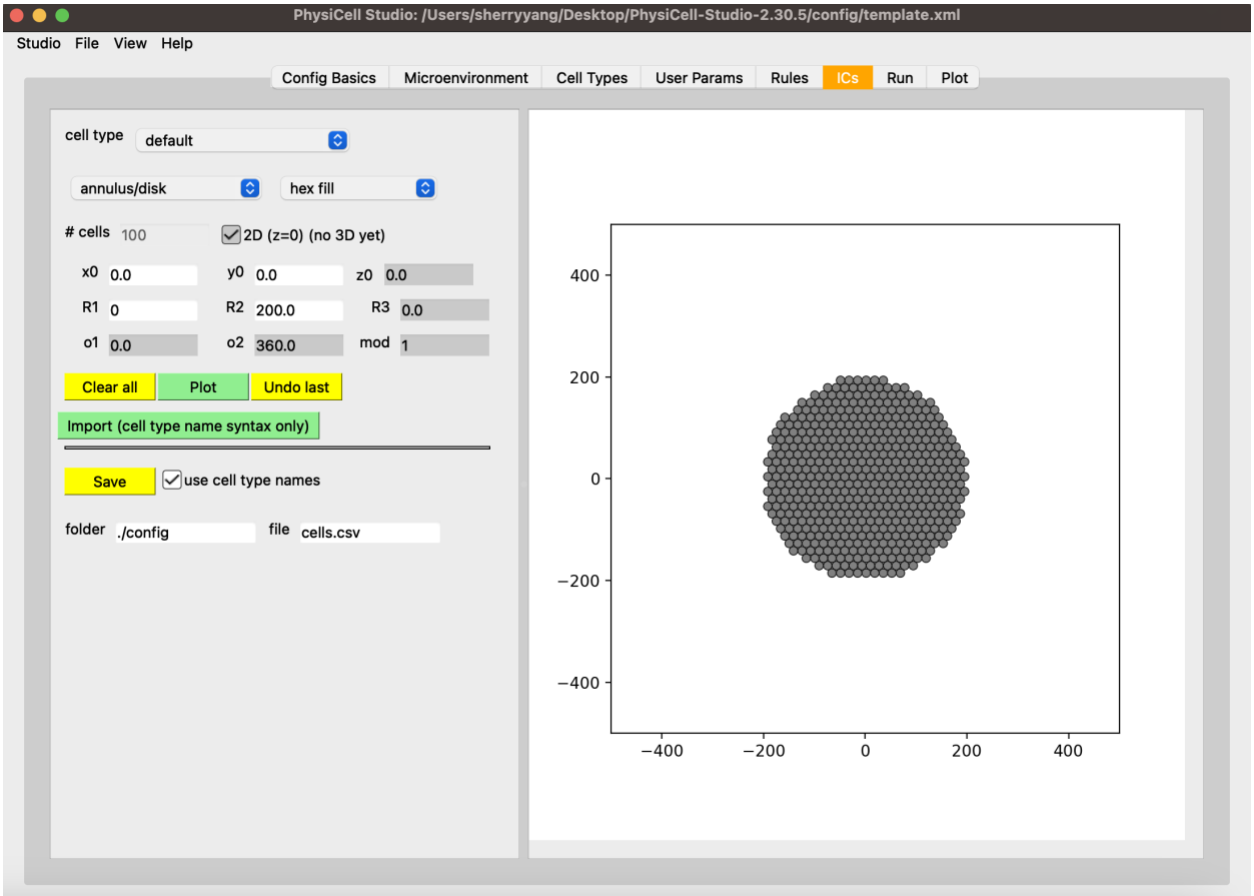

PhysiCell Studio: /Users/sherryyang/Desktop/PhysiCell-Studio-2.30.5rep/user_projects/studio_template/config/PhysiCell_settings.xml
Studio File View Help
Config Basics Microenvironment Cell Types User Params Rules ICs Run Plot
cell type effector T cell
annulus/disk random fill
cells 100 2D (z=0) (no 3D yet)
x0 0.0 y0 0.0 z0 0.0
R1 250 R2 500 R3 0.0
o1 0.0 o2 360.0 mod 1
Clear all Plot Undo last
Import (cell type name syntax only)
Save use cell type names
folder ./config file cells.csv
400
200
0
−200
−400
−400 −200 0 200 400

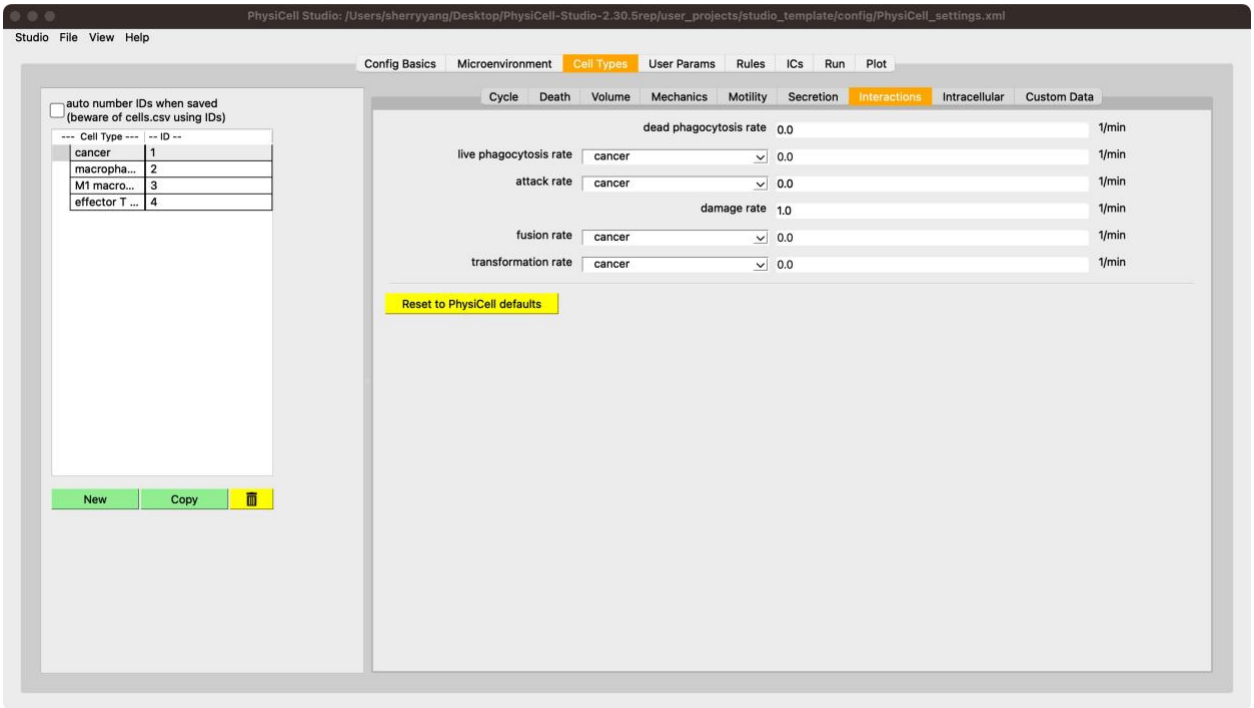


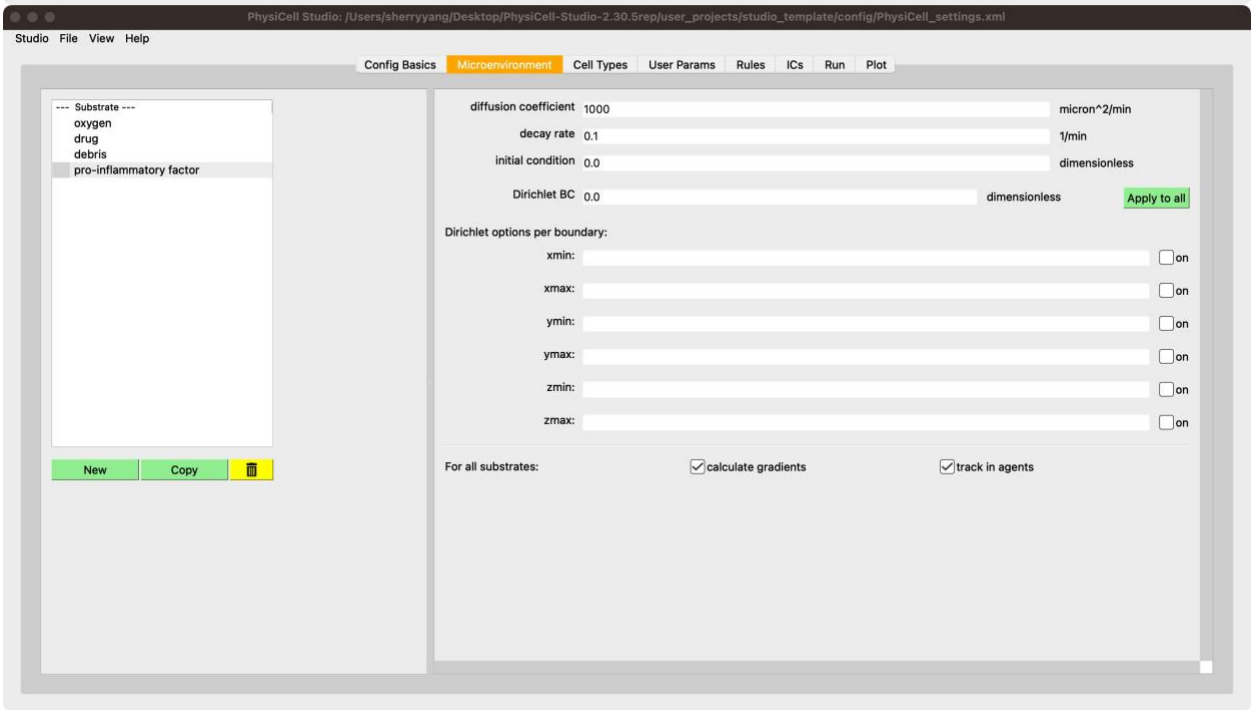


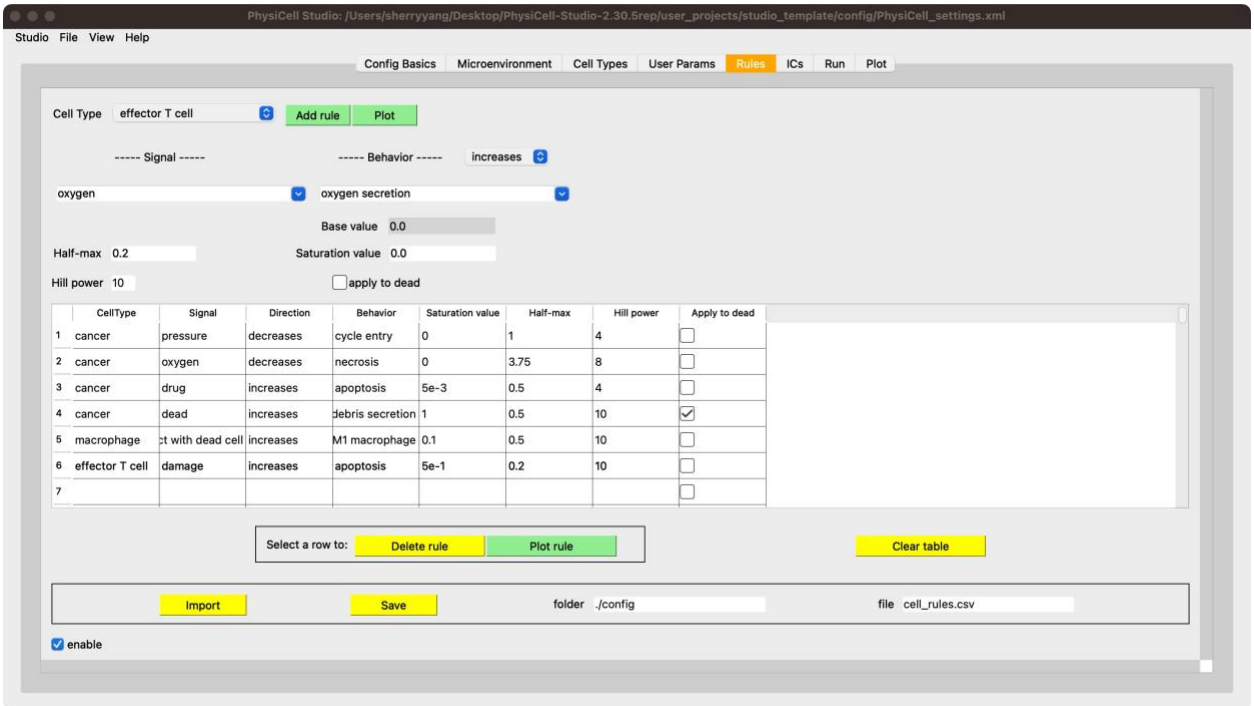
As shown in the figure, the greater R2, the more scattered are the effector T cells, this allows us to capture individual tumor microenvironment by changing these parameters

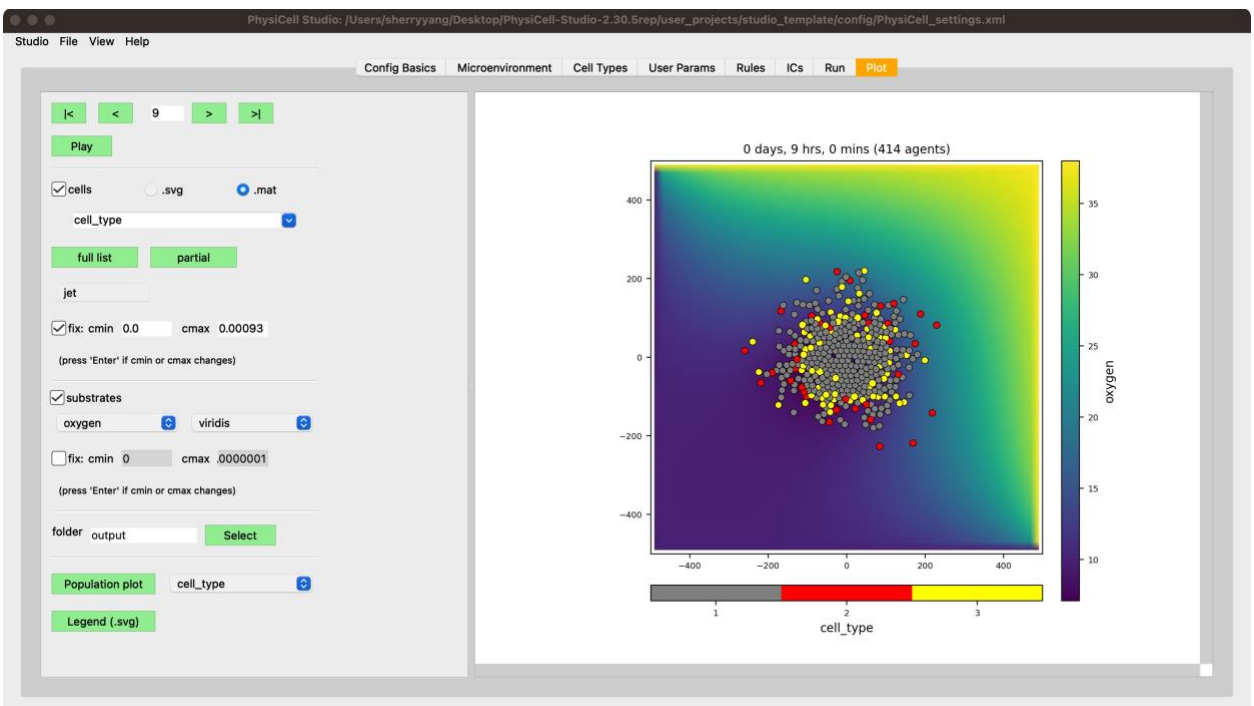
Simulation result

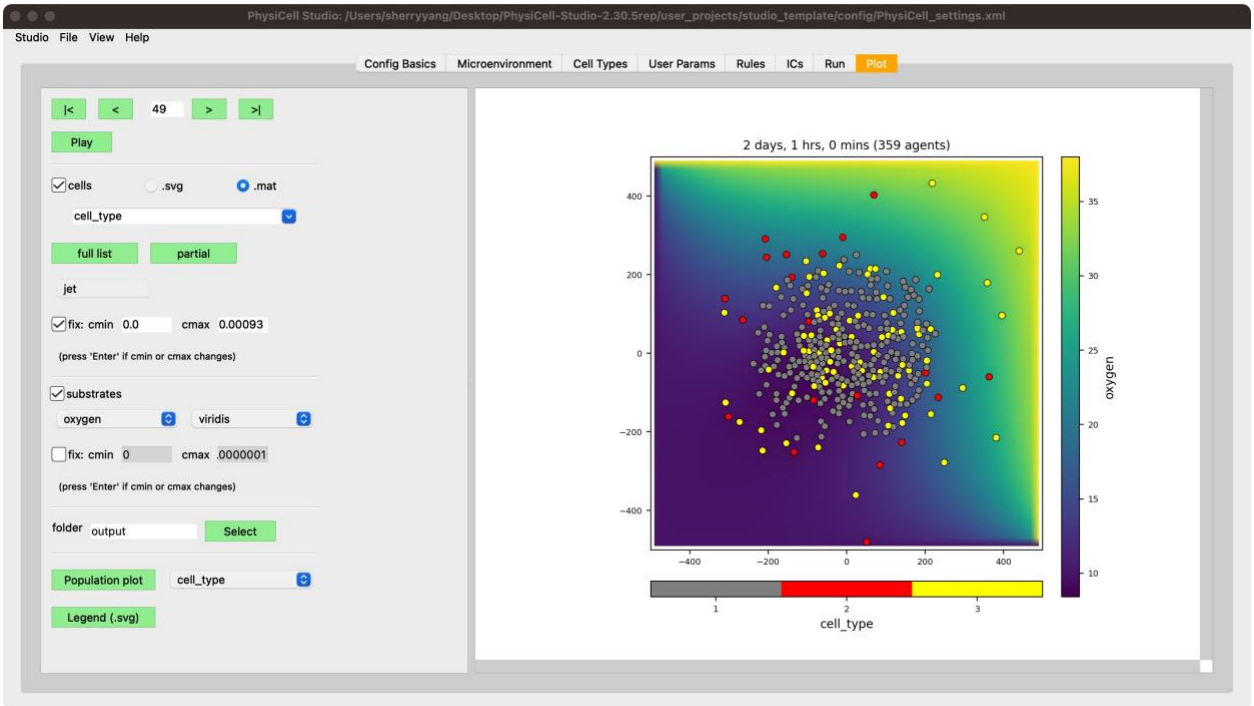



PhysiCell Studio: /Users/sherryyang/Desktop/PhysiCell-Studio-2.30.5rep/user_projects/studio_template/config/PhysiCell_settings.xml
Studio File View Help
Config Basics Microenvironment Cell Types User Params Rules ICs Run Plot
auto number IDs when saved (beware of cells.csv using IDs)
--- Cell Type --- -- ID --
cancer 1
macropha... 2
M1 macro... 3
effector T ... 4
New Copy
Cycle Death Volume Mechanics Motility Secretion Interactions Intracellular Custom Data
dead phagocytosis rate 0.0 1/min
live phagocytosis rate cancer 0.0 1/min
attack rate cancer 0.0 1/min
damage rate 1.0 1/min
fusion rate cancer 0.0 1/min
transformation rate cancer 0.0 1/min
Reset to PhysiCell defaults


PhysiCell Studio: /Users/sherryyang/Desktop/PhysiCell-Studio-2.30.5rep/user_projects/studio_template/config/PhysiCell_settings.xml
Studio File View Help
Config Basics Microenvironment Cell Types User Params Rules ICs Run Plot
--- Substrate ---
oxygen
drug
debris
pro-inflammatory factor
New Copy
diffusion coefficient 1000 micron^2/min
decay rate 0.1 1/min
initial condition 0.0 dimensionless
Dirichlet BC 0.0 dimensionless
Apply to all
Dirichlet options per boundary:
xmin: on
xmax: on
ymin: on
ymax: on
zmin: on
zmax: on
For all substrates: calculate gradients track in agents


PhysiCell Studio: /Users/sherryyang/Desktop/PhysiCell-Studio-2.30.5rep/user_projects/studio_template/config/PhysiCell_settings.xml
Studio File View Help
Config Basics Microenvironment Cell Types User Params Rules ICs Run Plot
Cell Type effector T cell
Add rule
Plot
----- Signal -----
----- Behavior -----
increases
oxygen
oxygen secretion
Base value 0.0
Half-max 0.2
Saturation value 0.0
Hill power 10
apply to dead
CellType Signal Direction Behavior Saturation value Half-max Hill power Apply to dead
1 cancer pressure decreases cycle entry 0 1 4
2 cancer oxygen decreases necrosis 0 3.75 8
3 cancer drug increases apoptosis 5e-3 0.5 4
4 cancer dead increases debris secretion 1 0.5 10
5 macrophage ct with dead cell increases M1 macrophage 0.1 0.5 10
6 effector T cell damage increases apoptosis 5e-1 0.2 10
7
Select a row to:
Delete rule
Plot rule
Clear table
Import
Save
folder ./config
file cell_rules.csv
enable


PhysiCell Studio: /Users/sherryyang/Desktop/PhysiCell-Studio-2.30.5rep/user_projects/studio_template/config/PhysiCell_settings.xml
Studio File View Help
Config Basics Microenvironment Cell Types User Params Rules ICs Run Plot
|< < 9 > >|
Play
cells .svg .mat
cell_type
full list
partial
jet
fix: cmin 0.0 cmax 0.00093
(press 'Enter' if cmin or cmax changes)
substrates
oxygen
viridis
fix: cmin 0 cmax .0000001
(press 'Enter' if cmin or cmax changes)
folder output
Select
Population plot
cell_type
Legend (.svg)
0 days, 9 hrs, 0 mins (414 agents)
oxygen
cell_type



According to simulation, over time, the immune cells from initial condition go from surrounding the cancer cells to merge into the center of tumor clumps to then roam and scatter around